IT5009
MULTIMEDIA TECHNOLOGIES

HYPERMEDIA CONCEPTS

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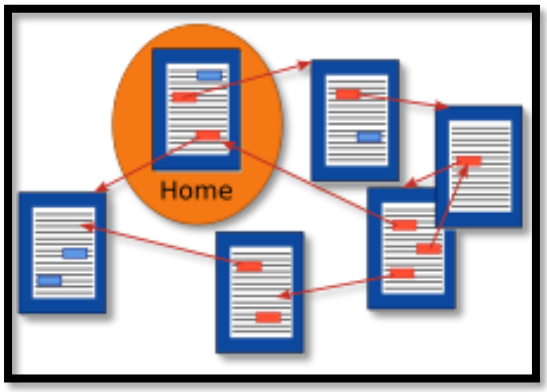
Hypertext

Hypertext is a text displayed on an electronic medium that contains links to other text within the same document or external files containing separate documents. It is the underlying structure which defines navigational pathways over the Internet, specifically the World Wide Web.



Hypertext documents are interconnected by hyperlinks, which are typically activated by a mouse click, keypress set, or screen touch.

The term "hypertext" is often used where the term "hypermedia" might seem appropriate.

In 1992, author Ted Nelson – who coined both terms in 1963 – wrote:

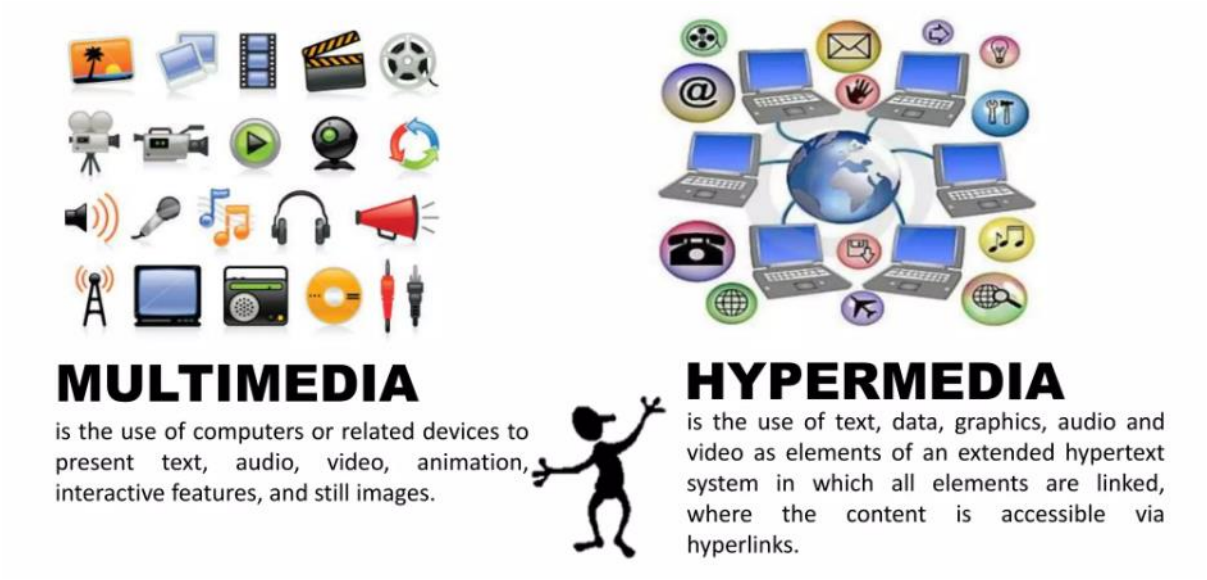
> By now the word "hypertext" has become generally accepted for branching and responding text, but the corresponding word "hypermedia", meaning complexes of branching and responding graphics, movies and sound – as well as text – is much less used. Instead, they use the strange term "interactive multimedia": this is four syllables longer, and does not express the idea of extending hypertext.
>
> – Nelson, *Literary Machines*, 1992

Hypermedia

Hypermedia is the logical extension of hypertext, where non-textual media like audio, video, image, animation, graphics, etc. can be used to create hyperlinks. It can be created by any programming tool capable of creating links from internal variables and nodes of one application to external files.

Hypermedia allows links to be embedded in multimedia elements like images and videos. You can tell if something is hypermedia by hovering the mouse cursor over the image or video – if the element is hypermedia, the cursor changes, usually into a small hand.

Although the Internet is the best example of the use of hypermedia, there is a lot of software that makes use of both hypermedia and hypertext. A lot of word processing, spreadsheet and presentation software like Microsoft Office allow hypermedia and hypertext to be embedded into the documents created. For example, in Microsoft Word, users can add hyperlinks to any word and even add links to pictures. Microsoft PowerPoint has the same feature for hypermedia.



Markup

Markup is everything in a document that is not content. It refers to a sequence of characters that you insert at certain places in a text or word-processing file to indicate how the file should look like when it is printed or displayed, or to describe the document's logical structure.

Markup indicators are called tags.

There are two types of markups:

1) Procedural Markup: Unique to a specific software package such as Microsoft Word. Each has its own set of markup codes that make sense only to itself. This markup usually takes the form of formatting codes that are mixed in with the text of the document. For example, <Ctrl>+B for making text bold in MS-Word.

2) Descriptive Markup: Its basic concept is that it should not be unique to specific software. It deals with a description stating the purpose of the change required, which should be recognised by different application software. Each of these applications can then product their own corresponding codes to implement the change. For example, a tag is used for making text bold in HTML pages. It is interpreted by different browsers which product the actual change in their own ways.

Some well-known markup languages are:

1) Hypertext Markup Language
2) Extensible Markup Language
3) Chemical Markup Language
4) Graph Exchange Language
5) Java Speech Markup Language
6) Mathematical Markup Language
7) Music Markup Language
8) Scalable Vector Graphics
9) Speech Synthesis Markup Language
10) Synchronised Multimedia Integration Language
11) Voice Markup Language
12) Web Ontology Language

Hypertext Transfer Protocol

HTTP is a protocol or a set of rules for data communication in a hypermedia information system, and more specifically, for the World Wide Web. HTTP functions in a request-response manner in a client-server environment.

A client typically submits a request for viewing a web page stored on a web server. A response involves sending the requested information from the server to the client, along with the completion status information.

HTTP is a stateless protocol. That is, server doesn't retain information about a specific client. Instead, cookies which are stored in the local storage of the client system or device are used to retain information.

By default, HTTP connections are not secure. HTTP requests and responses are sent in plaintext, which means that anyone can read them. This lack of security poses a serious issue, especially in modern browsers were data privacy and protection are extremely import.

HTTP connections can be made secure by using Secure Sockets Layer. HTTP with SSL is called HTTPS, which is what we see being used in most websites these days. HTTPS requests and responses are encrypted, so only the client which makes the request, and the server which provides the response, can read the actual data.

The HTTP status code informs the user of the status of the request. All HTTP response status codes are separated into five classes or categories. The first digit

of the status code defines the class of response, while the last two digits do not have any classifying or categorization role.

There are five classes defined by the standard:

- *1xx informational response* – the request was received, continuing process
- *2xx successful* – the request was successfully received, understood, and accepted
- *3xx redirection* – further action needs to be taken in order to complete the request
- *4xx client error* – the request contains bad syntax or cannot be fulfilled
- *5xx server error* – the server failed to fulfill an apparently valid request

1xx Information Exchange	100	Continue	The server has received the request headers and the client should proceed to send the request body.
	101	Switching Protocols	The requester has asked the server to switch protocols and the server has agreed to do so.
2xx Success	200	OK	Standard response for successful HTTP requests.
	201	Created	The request has been fulfilled, resulting in the creation of a new resource.
	202	Accepted	The request has been accepted for processing.
	204	No Content	The server processed the request, but is not returning any content.
3xx Redirection	304	Not Modified	Indicates that the resource has not been modified. In such a case, there is no need to retransmit the resource since the client still has a previously-downloaded copy.
	305	Use Proxy	The requested resource is available only through a proxy.

	307	Temporary Redirect	In this case, the request should be repeated with another URI; however, future requests should still use the original URI.
	308	Permanent Redirect	This and all future requests should be directed to the given URI.
4xx Client Errors	400	Bad Request	The server cannot or will not process the request due to an apparent client error.
	401	Unauthorised	Authentication is required but has failed or has not yet been provided.
	403	Forbidden	The request contained valid data and was understood by the server, but the server is refusing action.
	404	Not Found	The requested resource could not be found.
	408	Request Timeout	The server timed out waiting for the request.
	414	URI Too Long	The URI provided was too long for the server to process.
5xx Server Errors	500	Internal Server Error	A generic error message, given when an unexpected condition was encountered and no more specific message is suitable.
	502	Bad Gateway	The server was acting as a gateway or proxy and received an invalid response from the upstream server.
	503	Service Unavailable	The server can’t handle the request. Generally, this is a temporary state.
	504	Gateway Timeout	The server was acting as a gateway or proxy and did not receive a timely response from the upstream server.

HTTP resides in the Application Layer, and can use either the reliable TCP or the unreliable UDP of the Transport Layer for host-to-host data transfer. HTTP resources are identified using a URL – Uniform Resource Locator - which is embedded within a HTML document to enable users to navigate from one document to another.

Major request methods are GET, used for information retrieval, and POST, to submit information to a server using a form.

HTTP is a stateless protocol, which implies that the server doesn't retain information about a specific client. Thus, to track a user's progress, HTTP cookies are used, which is information served on the client machine and utilised by the server.

HTTP connections can be made secure by using SSL – Secure Sockets Layer – encryption. To differentiate it from standard HTTP connections, the protocol HTTPS is used.

Document Type Definition

A document written using a markup language is considered valid if it conforms to the rules laid out in a Document Type Definition. A DTD specification file is used to validate documents.

A DTD defines the valid building blocks of an XML document. It defines the document structure with a list of validated elements and attributes. A DTD can be declared inline inside an XML document, or as an external reference.

The main building blocks of a DTD are

Elements	represented by tags in HTML; use them to describe different parts of a data.
Attributes	provide additional information about tags, located inside opening tag of an element; use them to include metadata
Entities	set of characters having special meaning (eg: nbsp)
PCATA	parsed character data → text found between start and end tags, but without any entities.
CDATA	character data → not parsed by parser, represents tags inside text

The syntax for elements and attributes is as follows:

```
<!ELEMENT element-name (element-content)>
<!ATTLIST element-name attribute-name attribute-type default-value>
```

Here are some examples for HTML elements:

```
<!ELEMENT html (head, body)>
<!ELEMENT p (#PCDATA | p | ul | dl | table | h1|h2|h3)*>
```

Consider an anthology, which is a collection of poems, and each poem can consist of a single title and a number of stanzas, and each stanza can consist of a number of lines, and each line consists of parsed character data.

```
<anthology>
	<poem>
		<title>…</title>
		<stanza>
			<line>…</line>
			<line>…</line>
		</stanza>
		<stanza>
			<line>…</line>
			<line>…</line>
		</stanza>
	</poem>
	<poem>
		<title>…</title>
		<stanza>
			<line>…</line>
			<line>…</line>
		</stanza>
		<stanza>
			<line>…</line>
			<line>…</line>
		</stanza>
	</poem>
</anthology>
```

The DTD for the above would look something like:

<!ELEMENT anthology (poem+)>
<!ELEMENT poem (title, stanza+)>
<!ELEMENT stanza (line+)>
<!ELEMENT title (#PCDATA)>
<!ELEMENT line (#PCDATA)>

When it comes to attributes, consider an entry <square width="100"/>. The corresponding DTD declaration for this can be <!ATTLIST square width CDATA "0">.

```
<students>
    <student>
        <id>...</id>
        <name>...</name>
        <department>...</department>
        <grades>
            <subject>
                <name>...</name>
                <score>...</score>
            </subject>
        </grades>
    </student>
</students>
```

The DTD for the above would look something like:

<!ELEMENT students (student+)>
<!ELEMENT student (id, name, department, grades)>
<!ELEMENT id (#PCDATA)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT department (#PCDATA)>
<!ELEMENT grades (subject+)>
<!ELEMENT subject (name, score)>

The following occurrence indicators are used:

- A?: A can occur zero or one time
- A*: A can occur zero or more times
- A+: A can occur one or more times

The following group connectors are used:

- A|B: Both A and B are permitted in any order
- A, B: A must occur before B
- A&B: A and B must both occur once, but may do so in any order

Sometimes, there might be confusion as to when to use elements and when to use attributes. As a general rule, it is recommended to use elements to describe different parts of a data, because it enables the document to be searched for and identified more flexibly. However, the document might include metadata, which is information not part of the actual data but used to identify a particular document. It is preferable to include such metadata as attributes rather than elements.

Internal DTD is a type of DTD in that is written within the actual markup language document itself. Internal DTDs are suitable for smaller markup language documents where the complexity of the structure is not very high. It is easier to maintain and modify the internal DTD as it is part of the markup language document itself.

External DTD is a type of DTD that is located outside of the actual markup language document it describes. It can be stored in a separate file or accessed via a URL. By using an external DTD, multiple markup language documents can share the same set of rules and constraints, leading to more consistency and easier maintenance. External DTD can also be updated independently without having to modify the markup language documents themselves.

Standard Generalised Markup Language

SGML is an international standard published in 1966. It prescribes a standard format for embedding descriptive markup within a document. More importantly, and crucial to its real value and power, SGML also specifies a standard method for describing the structure of a document.

SGML allows you to create documents that are independent of any specific hardware or software. Since SGML documents conform to an international standard, they are portable. You can exchange them with users who have different systems.

SGML has 3 parts:

Structure: At the heart of an SGML application is the DTD. A document instance is a document whose content has been tagged in conformance with a particular DTD. A software program called a parser verifies that the document follows the rules of the DTD.

Content: Content is the information itself: alphanumeric text, images, graphics, audio, etc. The method for identifying the content's position within the DTD structure is called tagging. Creating an SGML document involves inserting tags around content. These tags mark the beginning and end of each part of the structure.

Style: SGML itself has nothing to do with setting standards for style, so most systems still rely on proprietary methods. Two efforts to develop standards-based style sheets have resulted in the development of FOSI and DSSSL. In 1996, the ISO approved the final draft of the Document Style Semantics and Specification Language (DSSSL) for SGML-based documents.

An SGML parser takes a DTD and generates from it a software system capable of validating any document invoking that DTD. It's output, at its simplest, is just "YES" (the document instance is valid), or "NO" (it is not).

A structured editor is a kind of intelligent word-processor. It can use information extracted from a processed DTD to prompt the user with information about which elements are required at different points in a

document, as the document is being created. It can also greatly simplify the task of preparing a document, for example, by inserting tags automatically.

A formatter operates on a tagged document instance to produce a printed form of it. Many typographic distinctions, such as the use of particular typefaces or sizes, are intimately related to structural distinctions, and formatters can thus usefully take advantage of descriptive markup.

The following is an appropriate DTD for SGML, for the abovementioned anthology example:

```
<!ELEMENT anthology  - -    (poem+)>
<!ELEMENT poem       - -    (title, stanza+)>
<!ELEMENT stanza     - O    (line+)>
<!ELEMENT title      - O    (#PCDATA)>
<!ELEMENT line       - O    (#PCDATA)>
```

The following is an appropriate DTD for SGML, for the abovementioned student table example:

```
<!ELEMENT            - -    students (student+)>
<!ELEMENT            - -    student (id, name, department, grades)>
<!ELEMENT            - O    id (#PCDATA)>
<!ELEMENT            - O    grades (subject+)>
<!ELEMENT            - O    subject (name, score)>
```

SGML specifies minimisation rules for the concerned elements. These rules determine whether or not start and end tags need to be present in every occurrence of the element concerned. They take the form of a pair of characters, separated by white space, the first of which relates to the start tag, and the second to the end tag.

In either case, a hyphen or a letter O must be given. The hyphen indicates that the tag must be present, and the letter O indicates that it is optional, that it may be omitted.

Dynamic HTML

DHTML, or Dynamic HTML, is a technology that combines HTML, CSS, JavaScript, and the Document Object Model (DOM) to create dynamic content.

HTML: HTML stands for Hypertext Markup Language and it is a client-side markup language. It is used to build the block of web pages. It is the predominant markup scripting language for the World Wide Web. It is interpreted line by line and executed.

HTML has been derived from SGML and has a fixed DTD. Its design objective is to display content of web pages on screen, and other than text, HTML also supports images, audio, video, interactive forms, tables, and multiple frames, each displaying a separate web page.

CSS: The abbreviation of CSS is Cascading Style Sheet. It helps in the styling of the web pages and helps in designing of the pages. HTML is sufficient to create a basic structure of a web document, but does not offer much facilities in fine tuning its appearance. For example, using HTML, one can’t precisely place an element on a web page, or fix sizes of textual characters accurately or overlap multiple media components.

CSS is an effort by W3C and web-browser vendors to remove these restrictions and allow for sophisticated sizing, positioning, and other appearance related manipulations of web page contents. One of the primary objectives of CSS is to separate content from style.

The CSS rules for DHTML can be modified at different levels using JS with event handlers which adds a significant amount of dynamism with very little code.

JavaScript: It is a cross-platform, object-based scripting language for client and server applications. JS enables creating applications that run over the internet. Using JS, one can create dynamic HTML pages that can process user input and maintain data using files and databases.

Although JS uses an object-oriented structure and dot notation, it is usually considered object-based instead of object-oriented, because it doesn’t define classes and works by using functions instead. The core language corresponds to

ECMA-262, the scripting language standardised by the European standards body European Computer Manufacturers Association (ECMA). The ECMA-262 standard is also approved by the ISO as ISO-16262.

JS is categorised into 3 different portions:

1) **Core JS:** comprises of the basic set of objects which makes up the foundation of the language, such as Array, Boolean, Date, String, and Math, and a core set of language elements such as operators, control structures, and statements.
2) **Client-side JS:** runs on the client and shares the same core language. Additionally uses certain features applicable only to the client side and used mainly for user interactions. Client side objects include Window, Document, Image, Form, Link, and Events.
3) **Server-side JS:** runs on the server and shares the same core language. Additionally extends the core language by supplying objects relevant to running JS on a server. For example, server-side extensions allow an application to communicate with a relational database, or perform file manipulations on a server. Server-side JS is compiled into bytecode executable files that are executed by a runtime engine on the web server.

By default, the term “JavaScript” refers to the client-side JS, including the core features. JS is enclosed within the SCRIPT element of an HTML markup.

Extensible HTML

XHTML is a family of markup languages generally regarded as extensions of HTML. One difference is that while HTML is based on SGML, XHTML is based on XML. XHTML provides a way for the developer to migrate to the XML world while maintaining compatibility with the older HTML platform. In 2007, the XHTML working group officially expired, as it was felt that HTML 5 was more suitable as the next generation HTML than XHTML 2. XHTML 5 was developed as part of HTML 5 specifications instead of a separate standard.

Document Object Model

DOM is an underlying platform to enable manipulation of web objects in a language-independent manner. A DOM is implemented within a web browser, and is used by languages like JS to change the content of a page dynamically.

DOM is a cross-platform and language-independent interface that treats an HTML or XML document as a tree structure wherein each node is an object representing a part of the document.

The DOM represents a document with a logical tree. Each branch of the tree ends in a node, and each node contains objects. DOM methods allow programmatic access to the tree; with them one can change the structure, style or content of a document. Nodes can have event handlers (also known as event listeners) attached to them. Once an event is triggered, the event handlers get executed.

The principal standardization of the DOM was handled by the World Wide Web Consortium (W3C), which last developed a recommendation in 2004. The Web Hypertext Application Technology Working Group (WHATWG) took over the development of the standard, publishing it as a living document. The W3C now publishes stable snapshots of the WHATWG standard.

The DOM tree can be manipulated using JavaScript or other programming languages. Common tasks include

- navigating the tree
- adding nodes
- removing nodes
- modifying nodes
- getting the properties of nodes
- setting the properties of nodes.

The DOM API provides a set of methods and properties to perform these operations, such as

- getElementById
- createElement
- appendChild
- innerHTML.

An HTML document with the following structure:

```
<html>
	<head>
		<title>My Website</title>
	</head>
	<body>
		<h1>Welcome to DOM</h1>
		<p>This is my website.</p>
	</body>
</html>
```

will be represented in the DOM tree as:

```
- Document (root)
	- html
		- head
			- title
				- "My Website"
		- body
			- h1
				- "Welcome"
			- p
				- "This is my website."
```

Attributes of an element are represented as properties of the element node in the DOM tree. For example, an element with the following HTML:

```
<a href="https://example.com">Link</a>
```

will be represented in the DOM tree as:

```
- a
	- href: "https://example.com"
	- "Link"
```

An XML document detailing a student schema with the following structure:

```
<student>
        <id>2022506001</id>
        <name>K Mohamed Jamal Basha </name>
        <department>IT</department>
        <grades>
                <subject>
                        <name>MMT</name>
                        <score>100</score>
                </subject>
        </grades>
</student>
```

will be represented in the DOM tree as:

```
- …
        - student
                - id
                        - "2022506001"
                - name
                        - "K Mohamed Jamal Basha“
                - department
                        - "IT"
                - grades
                        - subject
                                - name
                                        - "MMT"
                                - score
                                        - 100
```

Extensible Markup Language

XML is a markup language and file format for storing, transmitting, and reconstructing data. It defines a set of rules for encoding documents in a format that is both human-readable and machine-readable.

While HTML is concerned with the formatting and appearance of data, and focuses on how the content on a page is displayed to the user, XML is concerned with how to structure and store the data so that it can subsequently be searched for and possibly transported from one place to another.

Essentially, XML does not do anything. It is just information wrapped in tags. XML does not have standard tags. Rather, it enables the user to create customised tags to suit the document being stored. It allows one to create new languages by creating a new set of tags to describe specific type of data.

The name “extensible” comes from the fact that newer tags can be incorporated within the document without modifying the earlier tags, and the earlier application will continue to work.

XML with the correct syntax is called **well formed**. The rules for well-formed documents are

1) there must be a root element.
2) all elements must have closing tags.
3) tags are case-sensitive
4) elements must be properly nested
5) attribute values must be within quotes

An XML parser checks to see whether the syntax of an XML document is correct. For example, a parser will display an error if the start tag name doesn’t match with the end tag name.

XML which has been validated against a DTD is called **valid** XML. An XML DTD defines the structure of an XML document. If the parser gives an OK for a document, the document can then be validated against a DTD. The validation procedure checks to see whether the elements match with their definitions in the DTD. For example, if the element name is written incorrectly, the parser will

not generate an error, but the XML validator checks the name with the DTD and would generate an error.

W3C defined an alternative to a DTD called an XML Schema in 2001. Like a DTD, an XML Schema also defines the structure of an XML document.

One of the greatest advantages of XML Schema over DTDs is that schemas support datatypes, which makes it easier to define data patterns and check the validity of data.

Another advantage is that the schema itself is written in XML, due to which it has support for the XML DOM and can be parsed with an XML parser. An XML Schema external file has an extension XSD.

```
<xs:schema>
<xs:element name= "note ">
        <xs:complexType>
                <xs:sequence>
                        <xs:element name= "to " type= "xs:string "/>
                        <xs:element name= "from " type= "xs:string "/>
                        <xs:element name= "body" type= "xs:string "/>
                </xs:sequence>
        </xs:complexType>
</xs:element>
</xs:schema>
```

Display information can be added to an XML document using CSS for a customised look.

Scalable Vector Graphics

SVG is a set of XML-based specifications for describing 2D vector graphics. The advantages of SVG include the fact that they being vector formats do not suffer from pixelation due to zooming. Being XML, they can be edited using a text editor and can be indexed and searched for.

SVG files are saved using the .SVG extension.

SVG supports interactivity, animation, and rich graphical capabilities, making it suitable for both web and print applications.

SVG is an ideal choice for high-resolution displays and responsive web design. SVG images are made up of paths, shapes, and text, which are defined by mathematical equations. These equations allow SVG images to be scaled up or down without losing any quality, making them resolution-independent.

One of the major advantages of using SVG images is their small file size. Since the images are defined by mathematical equations rather than pixels, the file size is often much smaller than that of a raster image, such as a PNG. This is especially beneficial for web developers looking to optimize the performance of their websites.

Another advantage of SVG images is their ability to be easily manipulated using CSS and JavaScript. This means that you can change the appearance of an SVG image on your website simply by adjusting some code, without needing to create a new image file. This can be a huge time-saver for web designers and developers.

If you're working with simple graphics, icons, or logos, SVG is the optimal choice due to its smaller file sizes and ability to be scaled without losing quality.

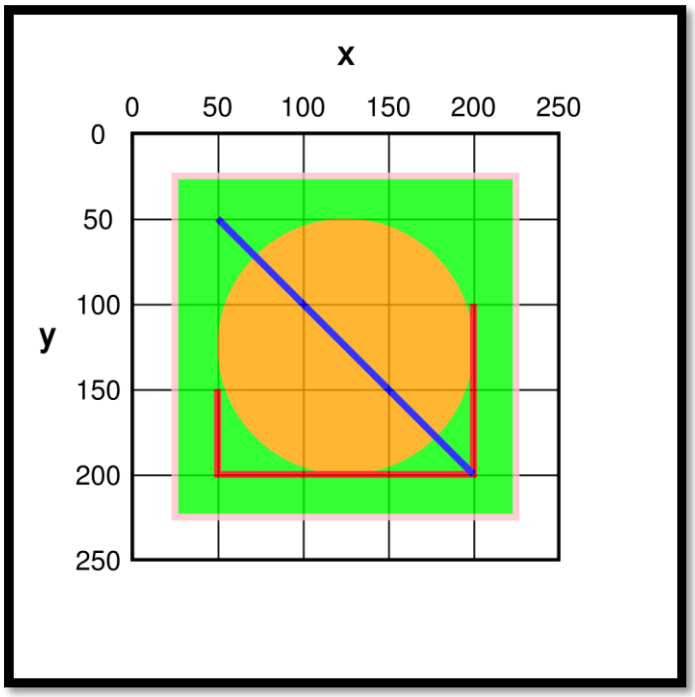
SVG images can be compressed with the gzip algorithm, resulting in SVGZ files that are typically 20–50% smaller than the original. SVG also supports metadata, enabling better indexing, searching, and retrieval of SVG content.

As a document format, similar to HTML documents, SVG can host scripts or CSS. This is an issue when an attacker can upload a SVG file to a website, such

as a profile picture, and the file is treated as a normal picture but contains malicious content.

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN"
"http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd">
<svg width="391" height="391" viewBox="-70.5 -70.5 391 391"
xmlns="http://www.w3.org/2000/svg"
xmlns:xlink="http://www.w3.org/1999/xlink">
<rect fill="#fff" stroke="#000" x="-70" y="-70" width="390" height="390"/>
<g opacity="0.8">
	<rect x="25" y="25" width="200" height="200" fill="lime" stroke-
width="4" stroke="pink" />
	<circle cx="125" cy="125" r="75" fill="orange" />
	<polyline points="50,150 50,200 200,200 200,100" stroke="red"
stroke-width="4" fill="none" />
	<line x1="50" y1="50" x2="200" y2="200" stroke="blue" stroke-
width="4" />
</g>
</svg>
```

The above code will produce the coloured shapes as shown in the image below, excluding the grid and labels:



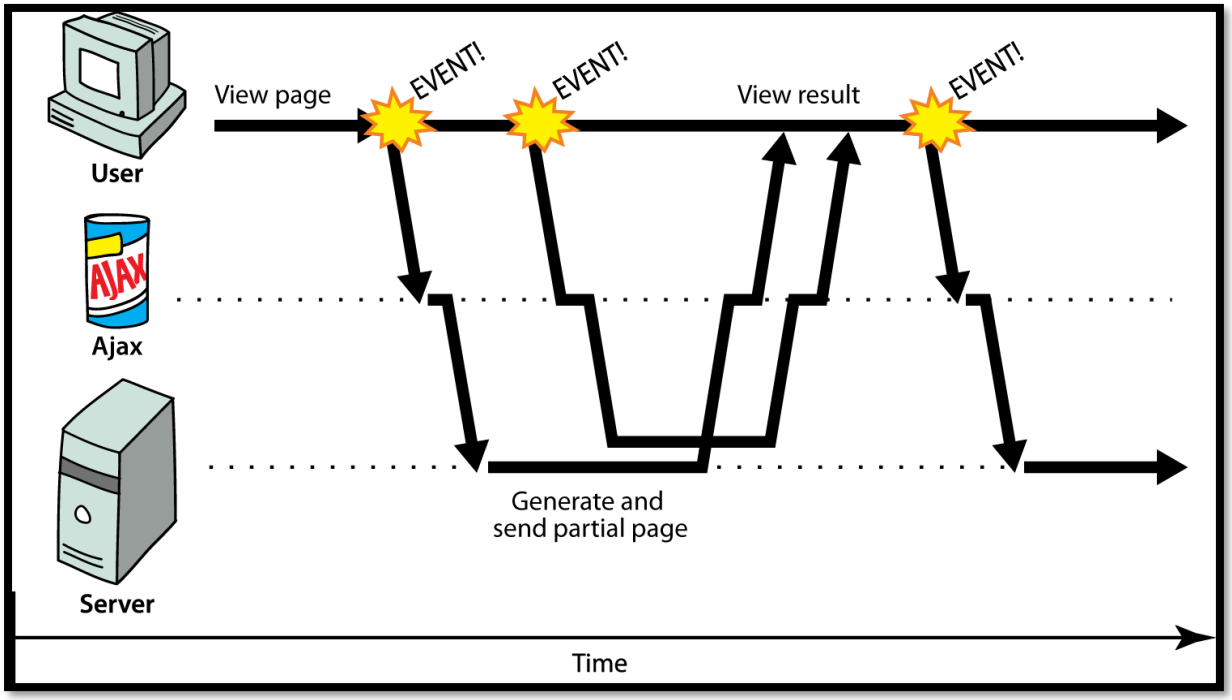
Asynchronous JavaScript and XML

AJAX is a set of web development techniques that uses various web technologies on the client-side to create asynchronous web applications. Asynchronous applications are those which retrieve data from the server asynchronously in the background, without disturbing the rest of the page content.

AJAX is not a separate technology, but rather a combination of existing technologies.

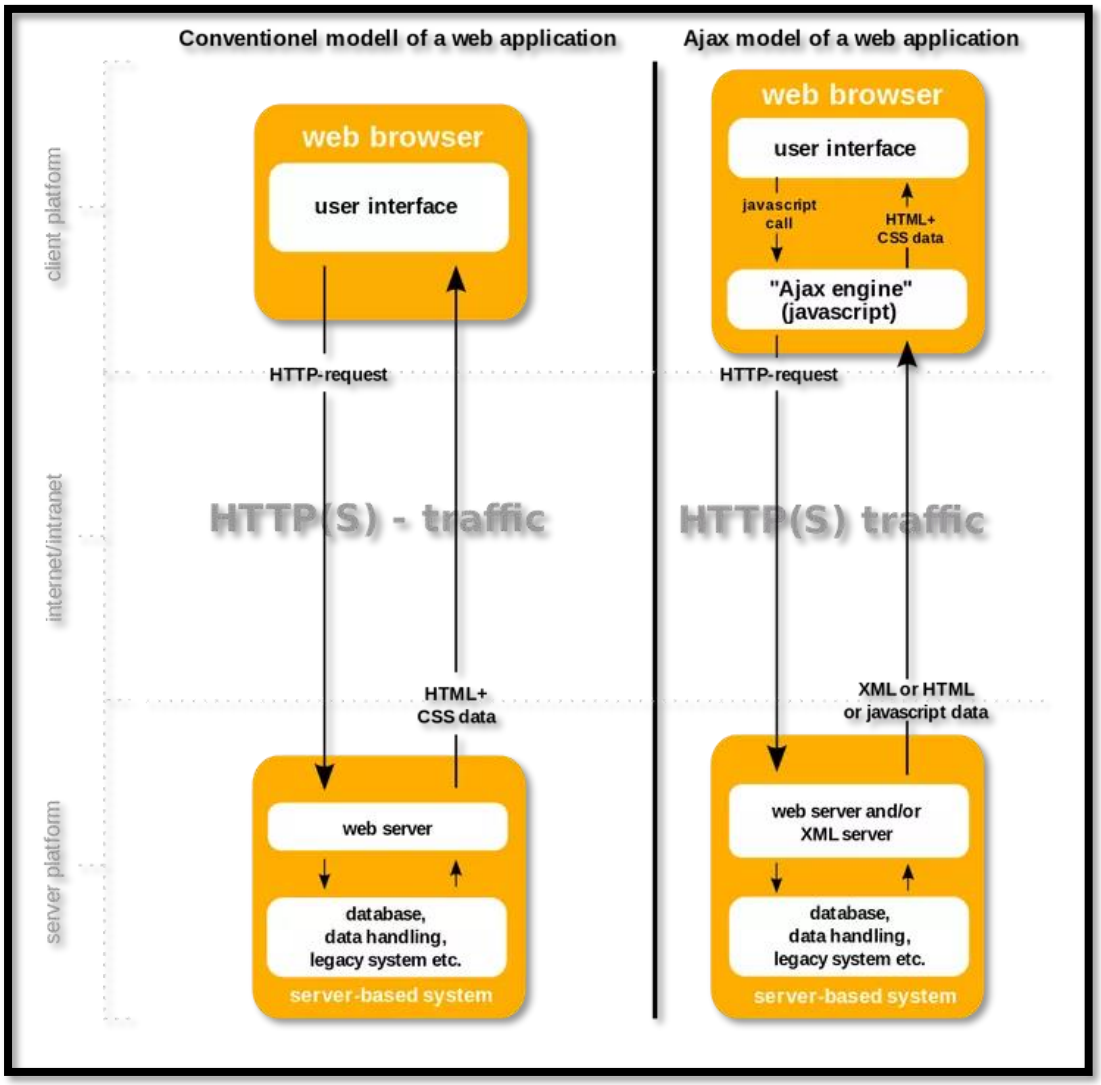
In the article that coined the term Ajax,[1][3] Jesse James Garrett explained that the following technologies are incorporated:

- HTML (or XHTML) and CSS for presentation
- The Document Object Model (DOM) for dynamic display of and interaction with data
- JSON or XML for the interchange of data, and XSLT for XML manipulation
- The XMLHttpRequest object for asynchronous communication
- JavaScript to bring these technologies together



A variety of popular JavaScript libraries, including JQuery, include abstractions to assist in executing AJAX requests.

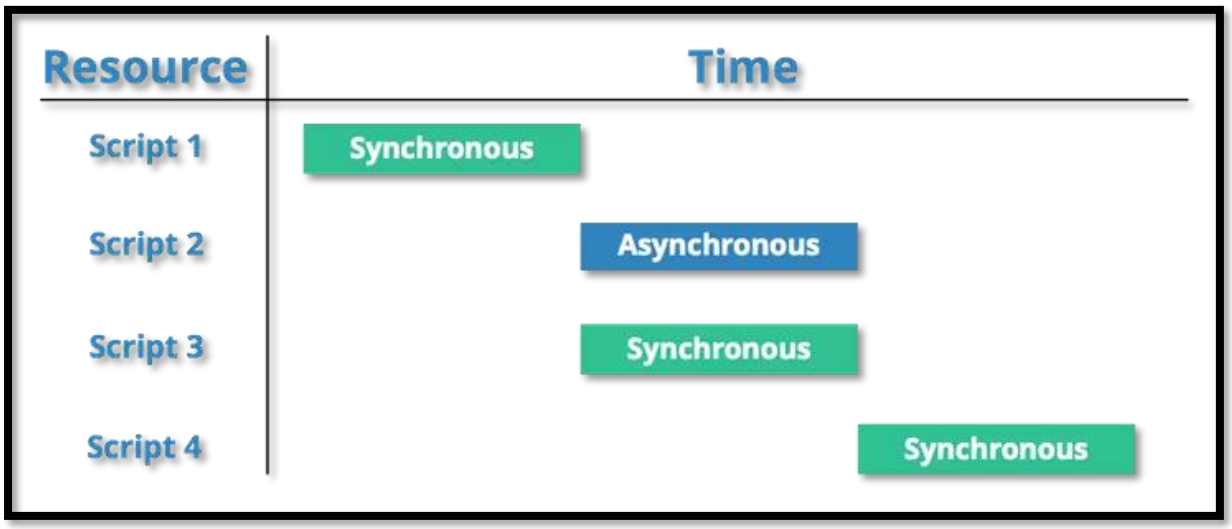
Ajax offers several benefits that can significantly enhance web application performance and user experience. By reducing server traffic and improving speed, Ajax plays a crucial role in modern web development. One key advantage of Ajax is its capacity to render web applications without requiring data retrieval, resulting in reduced server traffic. This optimization minimizes response times on both the server and client sides, eliminating the need for users to endure loading screens.



When it comes to loading scripts, traditionally scripts were loading using the synchronous method. This was more or less of a "first-come first-served" process whereas if during the page rendering process, the HTML came across a script tag then it would load that script tag in sequence, thus blocking the

parsing of HTML. Once the script was fully loaded, the HTML would continue to parse and load the rest of the web page.

As we can see by the image below, there are 3 synchronous scripts being loaded one after another. However, there is also an asynchronous script which can be loaded at the same time as the synchronous script which makes for a much more efficient loading process.



Multipurpose Internet Mail Extensions

MIME is an identifier for file formats on the Internet. It was first defined in RFC 2045 for use with email sent through SMTP, where it worked by adding additional information to the email message header, indicating the type of file being transmitted and how it should be handled by the recipient's email client. Since then, it has expanded to other protocols such as HTTP and RTP.

In the HTTP for the World Wide Web, servers insert a MIME header field at the beginning of any Web transmission. Clients use the content type or media type header to select an appropriate viewer application for the type of data indicated.

A MIME header contains multiple subparts. These include:

MIME-Version

This header field indicates the message is MIME-formatted. Its value is typically "1.0."

Content-Type

It defines media type as a composition of two parts: a type and a subtype, and is written using the form type/subtype.

- Type: General category of the data type (e.g., video or text)
- Subtype: Exact kind of data the MIME type represents.

MIME types are case-insensitive. An optional parameter can be added to the MIME type to provide additional details. So, if the parameter is "charset", it will specify the character set used for the data characters. If this parameter is not specified, MIME will use the default ASCII character set.

Text	**Image**	**Audio**	**Application**	**Video**
text/plain	image/gif	audio/mpeg	application/javascript	video/mpeg
text/html	image/jpeg	audio/mp4	application/pdf	video/mp4
text/css	image/png	audio/ogg	application/zip	video/ogg
text/xml	image/tiff	audio/wave	application/powerpoint	video/quicktime

Content-Disposition

This header specifies the presentation style of email messages. The disposition may be:

- Inline: Automatically displayed when the message is displayed
- Attachment: Not displayed automatically but rather requires some user action (i.e. download)

Content-Transfer-Encoding: This header indicates whether a binary-to-text encoding scheme has been used in addition to the original encoding specified in the content-type header.

Synchronised Multimedia Integration Language

SMIL (pronounced ‘smile’) is an XML-based W3C recommendation for describing multimedia presentations. It defines markup elements for timing, layout, animations, and transitions.

SMIL documents are like HTML documents in structure. Both are typically divided into an optional <head> section and a required <body> section. The body section contains the timing information and is generally composed of combinations of the three main tags <seq>, <par>, and <excl>. But SMIL’s focal core is synchronizing zones with multimedia content, while HTML spotlights structuring and linking text and multimedia content.

It divides the different media components into separate files and streams and then displaying them together on the user's computer as if they were a single multimedia stream. For example, SMIL is commonly used for delivering captioned text and/or audio description, synchronized with an accompanying video file.

Below is a snippet of an SMIL file defining a slide change sequence.

```
<slidechange>
        <slide id="1" imagefile="Slide001.bmp" st="0.0000" et="5.0400" type="normal" />
        <slide id="2" imagefile="Slide002.bmp" st="5.0400" et="5.5200" type="skip" />
        <slide id="3" imagefile="Slide003.bmp" st="5.5200" et="5.9600" type="skip" />
        <slide id="4" imagefile="Slide004.bmp" st="5.9600" et="6.2800" type="skip" />
        <slide id="5" imagefile="Slide005.bmp" st="6.2800" et="7.2400" type="skip" />
        <slide id="6" imagefile="Slide006.bmp" st="7.2400" et="12.8400" type="normal" />
        <slide id="7" imagefile="Slide007.bmp" st="12.8400" et="12.9600" type="skip" />
        <slide id="8" imagefile="Slide008.bmp" st="12.9600" et="13.1200" type="skip" />
        <slide id="9" imagefile="Slide009.bmp" st="13.1200" et="13.2800" type="skip" />
        <slide id="10" imagefile="Slide010.bmp" st="13.2800" et="13.4800" type="skip" />
        <slide id="11" imagefile="Slide011.bmp" st="13.4800" et="19.2000" type="normal" />
        <state TotalSlide="11" StablePeriod="3" UnstablePeriod="2" />
 </slidechange>
```

id: Unique identifier for the slide.
imagefile: The image file name for the slide.
st: When the slide appears.
et: When the slide disappears.
type: Defines if the slide is "normal" (displayed) or "skip" (ignored in the final playback).

Multimedia Messaging Service

MMS is a technique for sending messages containing multimedia content to and from mobile devices. MMS messages can contain text (formatted with different fonts, colours, and styles), images, audio, video, and animations.

MMS messages are first associated with a MIME type and then sent to the network provider's store, and from there, the message can be either directly sent to the specific mobile handset, or a control SMS is sent to the mobile device with a URL link to retrieve the message from the store. If the mobile device is not capable of handling MMS messages, a link can be sent to the handset, using which the message can be opened on a computer via a normal browser.

MMS is a natural evolution of SMS, which supports only text-based communication, but it is still limited in many ways.

Rich Communication Services, or RCS, is proposed as the modern alternative and solution. RCS has been designed as an industry open standard to provide improved capabilities over basic text messaging, based on the Internet Protocol (IP). Its development has also been supported by mobile network operators to regain their influence against individual OTT (over-the-top) chat apps and services such as WhatsApp, Telegram, and Signal.

Feature	**SMS**	**MMS**	**RCS**
Character Limit	160 characters	1600 characters	No
File Size Limit	No file support	5 MB	100 MB
Format	Text	Text and Multimedia	Text and Multimedia
Sent Via	Cellular	Cellular	Wi-Fi or Cellular
Security	No encryption	No E2EE	End-to-End Encrypted
Read Receipts	No	No	Yes
Typing Indicators	No	No	Yes
Group Chat Limit	No support	10 people	100 people

Internet Required	No	No	Yes
Reactions/Emojis	Limited (text-based)	Limited	Yes
Voice Messages	No	No	Yes
Video Calls	No	No	Yes
Backup Support	No	No	Cloud Backup

RCS can operate over cellular data, but it also has the potential to fall back on SMS/MMS in areas without a stable internet connection.

RCS is the Android standard and is the basis of Google Messages. Apple has its own version, called iMessage.

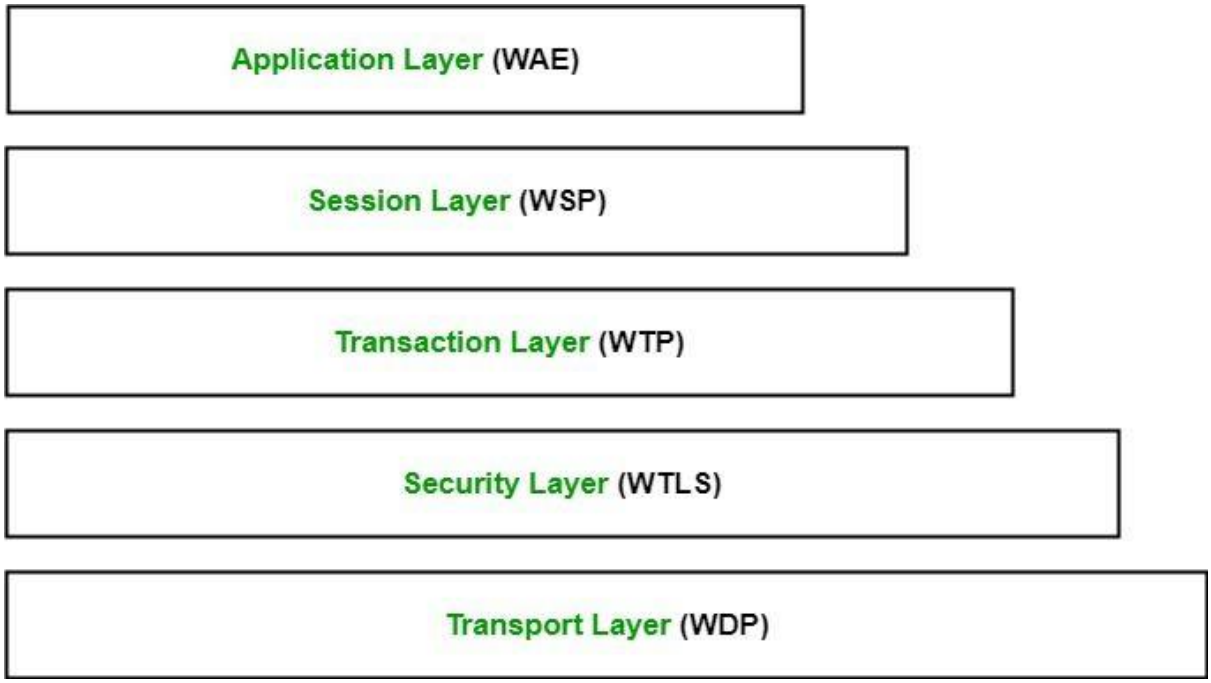
For many reasons, Google wanted Apple to drop iMessage in favour of RCS. According to Google, this would allow more fluid communications between Android and Apple users.

As a compromise, alongside iMessage, Apple has provided support for RCS as well in its software quite recently, meaning RCS is evolving to become a universal standard of sorts for advanced messaging.

RCS chats between Google Messages users are automatically upgraded to end-to-end encryption. This feature is not yet available for cross-platform conversations between Android and iOS devices. The GSMA, the governing body behind the RCS standard, is working to add interoperable end-to-end encryption for RCS chats between different platforms, but this feature is still in development and has not been fully implemented

Wireless Application Protocol

WAP is an open standard protocol suite to display internet content on mobile devices. It describes a complete protocol suite for mobile internet access. It allows content to be pushed to a mobile handset using minimal user intervention.



Wireless Datagram Protocol	Helps in transport of data in an unreliable way, similar to UDP.
Wireless Transport Layer Security	Provides public-key encryption based security, similar to SSL.
Wireless Transaction Protocol	Provides a reliable transport mechanism for transmitting data between the wireless device and the server.
Wireless Session Protocol	Responsible for maintaining a session between the wireless device and the server
Wireless Application Environment	Enables delivery of web-based content.

WAP played a significant role in the early days of mobile internet access and paved the way for the development of more advanced technologies that we use today. It has been largely replaced by more advanced protocols like HTML5 and native mobile apps.

WAP became obsolete because it was slow and limited, designed for early mobile phones with small screens and low bandwidth, making it incapable of handling modern, feature-rich websites.

As technology advanced, better mobile browsers like Chrome, Safari, and Firefox emerged, allowing smartphones to render full-fledged HTML, CSS, and JavaScript, eliminating the need for WAP. The adoption of HTML5, CSS3, and JavaScript, along with responsive web design, ensured that modern websites work seamlessly across all devices, removing the necessity for a separate WAP version.

Faster mobile networks such as 3G, 4G, and 5G further contributed to its decline by enabling high-speed internet access, making WAP's text-based pages redundant. The evolution of smartphones, particularly with iOS and Android, allowed users to browse the web without the restrictions of WAP-based sites.

Summary

Hypertext	Text with links to other text.
Hypermedia	Multimedia (text, images, videos) linked like hypertext.
Markup	Symbols/tags defining document structure.
HTTP	Protocol for web communication.
DTD	Defines rules for SGML/XML documents.
SGML	Parent standard of HTML/XML.
DHTML	HTML+CSS+JS for interactive web pages.
XHTML	Stricter version of HTML based on XML.
DOM	API for interacting with HTML/XML documents.
XML	Structured data format for information exchange.
SVG	XML-based format for vector graphics.
AJAX	Web technique for updating content dynamically.
MIME	Standard for defining file types on the web.
SMIL	Language for multimedia presentations.
MMS	Standard for sending multimedia messages.
WAP	Early protocol for mobile web access.